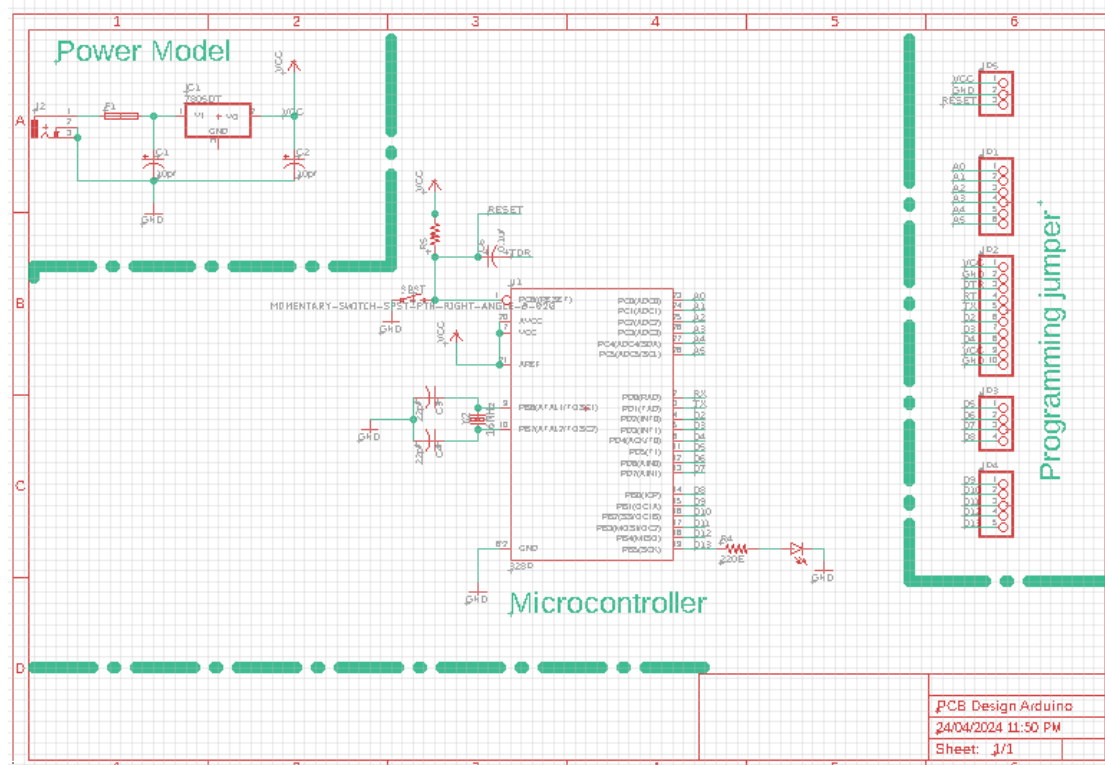


Lab Task: PCB design for Arduino board.

1. Power supply
1. Crystal oscillator
2. Microcontroller
3. LED
4. Pin header connection (Programming jumper)



Clock Generators: Electronic device required to generate clock for the digital system. oscillation is required to generate this clock, the processes to amplify the signal and feedback the signal back as input with 360° .

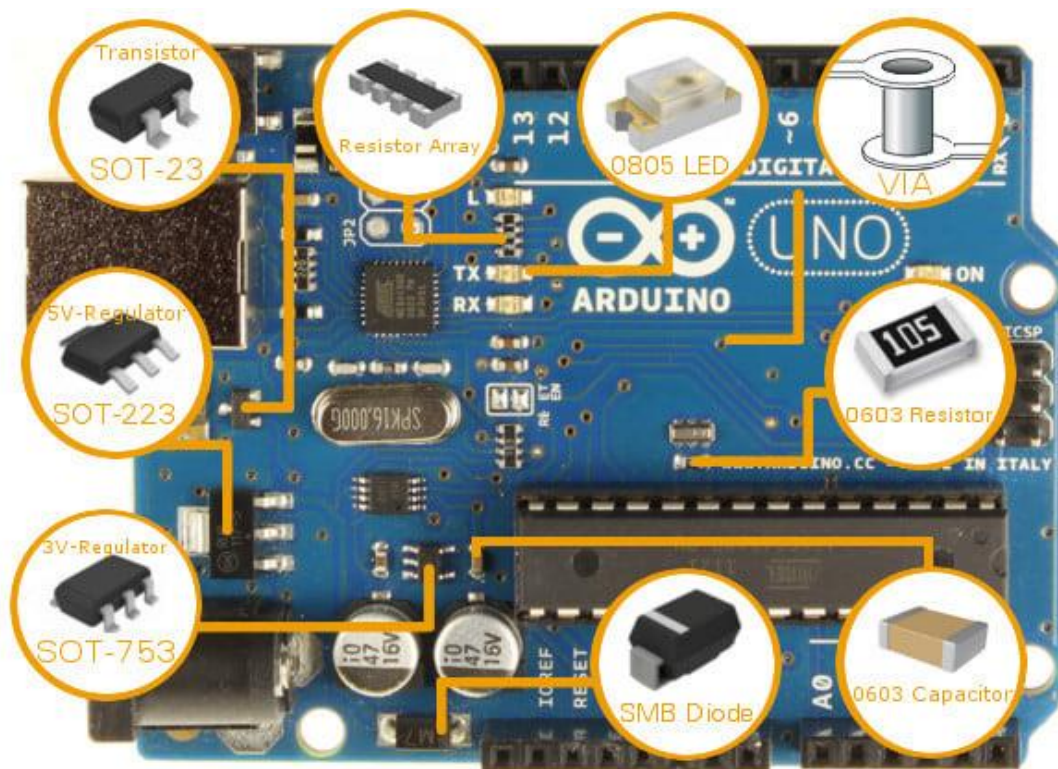
There are different ways to generate a clock and such as:

1. Crystal with additional caps and inverters to build a clock circuit.
2. Single oscillator IC and required power to obtain clock signal.
3. Programming Timer IC that used a few passives to set the frequency.

https://en.wikipedia.org/wiki/Crystal_oscillator

The core of all digital systems and computers since all digital computers systems are synchronous machines, they use a base clock signal and step through logic, program, states based on this clock.

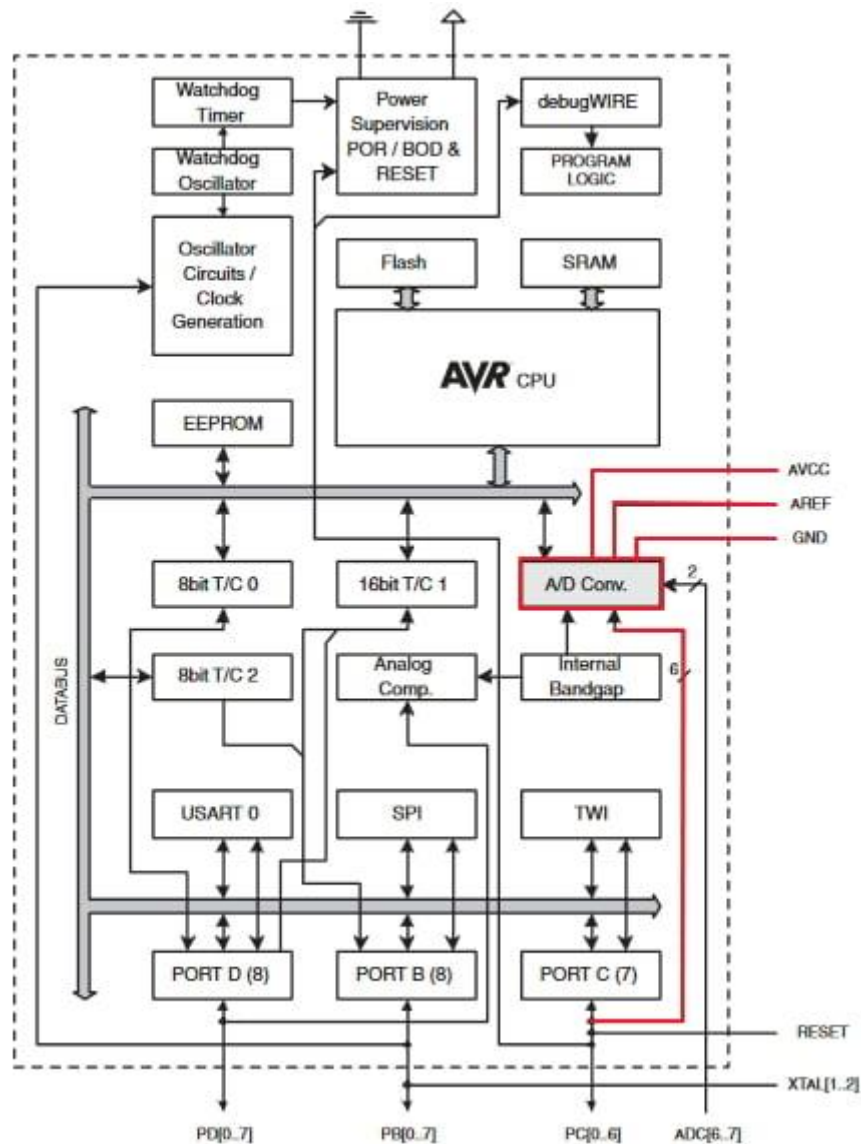
Arduino board:



Arduino History

<https://en.wikipedia.org/wiki/Arduino#:~:text=The%20Arduino%20project%20began%20in,environment%20using%20sensors%20and%20actuators.>

<https://www.arduino.cc/>



ATmega328 block diagram.

The Microcontroller

The ATmega328 microcontroller is the MCU used in Arduino UNO R3 as a main controller. ATmega328 is an MCU from the AVR family; it is an 8-bit device, which means that its data-bus architecture and internal registers are designed to handle 8 parallel data signals.

ATmega328 has three types of memory:

- **Flash memory:** 32KB nonvolatile memory. This is used for storing application, which explains why you don't need to upload

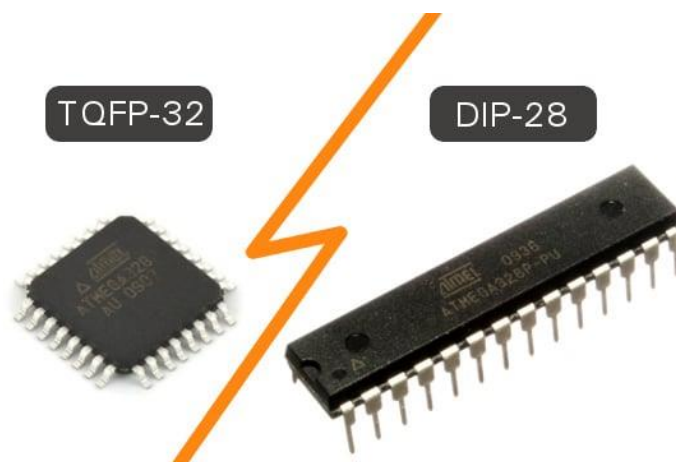
your application every time you unplug arduino from its power source.

- **SRAM memory:** 2KB volatile memory. This is used for storing variables used by the application while it's running.
- **EEPROM memory:** 1KB nonvolatile memory. This can be used to store data that must be available even after the board is powered down and then powered up again.

Packages:

This MCU is a DIP-28 package, which means that it has 28 pins in the dual in-line package. These pins include power and I/O pins. Most of the pins are multifunctional, which means that the same pin can be used in different modes based on how you configure it in the software. This reduces the necessary pin count, because the microcontroller does not require a separate pin for every function. It can also make your design more flexible, because one I/O connection can provide multiple types of functionalities.

Other packages of ATmega328 are available like TQFP-32 SMD package (Surface Mount Device).



Power:

The MCU accepts supply voltages from 1.8 to 5.5 V. However, there are restrictions on the operating frequency; for example, if you want to use the maximum clock frequency (20 MHz), you need a supply voltage of at least 4.5 V.

Digital I/O:

This MCU has three ports: PORTC, PORTB, and PORTD. All pins of these ports can be used for general-purpose digital I/O or for the alternate functions indicated in the pinout below. For example, PORTC pin0 to pin5 can be ADC inputs instead of digital I/O.

There are also some pins that can be configured as PWM output. These pins are marked with “~” on the Arduino board.

Note: The ATmega168 is almost identical to the ATmega328 and they are pin compatible. The difference is that the ATmega328 has more memory—32KB flash, 1KB EEPROM, and 2KB RAM compared to the ATmega168's 16KB flash, 512 bytes EEPROM, and 1KB RAM.

Atmega168 Pin Mapping

Arduino function						Arduino function
reset	(PCINT14/RESET) PC6	1	28	PC5 (ADC5/SCL/PCINT13)		analog input 5
digital pin 0 (RX)	(PCINT16/RXD) PD0	2	27	PC4 (ADC4/SDA/PCINT12)		analog input 4
digital pin 1 (TX)	(PCINT17/TXD) PD1	3	26	PC3 (ADC3/PCINT11)		analog input 3
digital pin 2	(PCINT18/INT0) PD2	4	25	PC2 (ADC2/PCINT10)		analog input 2
digital pin 3 (PWM)	(PCINT19/OC2B/INT1) PD3	5	24	PC1 (ADC1/PCINT9)		analog input 1
digital pin 4	(PCINT20/XCK/T0) PD4	6	23	PC0 (ADC0/PCINT8)		analog input 0
VCC	VCC	7	22	GND		GND
GND	GND	8	21	AREF		analog reference
crystal	(PCINT6/XTAL1/TOSC1) PB6	9	20	AVCC		VCC
crystal	(PCINT7/XTAL2/TOSC2) PB7	10	19	PB5 (SCK/PCINT5)		digital pin 13
digital pin 5 (PWM)	(PCINT21/OC0B/T1) PD5	11	18	PB4 (MISO/PCINT4)		digital pin 12
digital pin 6 (PWM)	(PCINT22/OC0A/AIN0) PD6	12	17	PB3 (MOSI/OC2A/PCINT3)		digital pin 11(PWM)
digital pin 7	(PCINT23/AIN1) PD7	13	16	PB2 (SS/OC1B/PCINT2)		digital pin 10 (PWM)
digital pin 8	(PCINT0/CLKO/ICP1) PB0	14	15	PB1 (OC1A/PCINT1)		digital pin 9 (PWM)

Digital Pins 11, 12 & 13 are used by the ICSP header for MOSI, MISO, SCK connections (Atmega168 pins 17, 18 & 19). Avoid low-impedance loads on these pins when using the ICSP header.